

ECOTEC_Online Student Risk Dataset

Codebook and Metadata Documentation

This codebook documents the final 18-variable public release, the operational definition and provenance of TARGET_RISK, documented public-release treatments and generalization, released financial-variable categories and values, and the analytical scope of the dataset.

Dataset Overview

Attribute	Final public release
Institution	Universidad ECOTEC, Ecuador
Educational modality	Online higher education
Records	12,632 unique students
Variables	18
Null cells	0
Duplicate rows	0
Unique PUBLIC_RECORD_ID	12,632
Restricted institutional research extract	12,632 records x 42 variables

Target Distribution

TARGET_RISK	n	%
0	691	5.47
1	11,941	94.53

Interpretive limitation. TARGET_RISK is a contemporaneous administrative proxy constructed from the same institutional snapshot as the released predictors. It is not a prospectively observed dropout or retention outcome.

Variable Dictionary - Part I

Variable	Description	Type	Missing	Unique
PUBLIC_RECORD_ID	Release-specific public record identifier replacing the restricted institutional student-record code. No public crosswalk to the original institutional identifier is released. It is intended only for within-release record referencing and is not intended or documented as a longitudinal person-level linkage key.	Numeric	0	12632
FACULTY	Academic faculty associated with the released student record.	Categorical/Text	0	1
DEGREE_PROGRAM	Academic degree program represented as a standardized public category.	Categorical/Text	0	29
CURRICULUM_YEAR	Curriculum or study-plan year associated with the degree program. It should not be interpreted as the student observation year or academic cohort.	Numeric	0	4
GENDER	Student gender represented as a standardized categorical variable.	Categorical/Text	0	2
AGE	Student age in completed years. Integer age is retained and is not generalized into age bands.	Numeric	0	51
MARITAL_STATUS	Student marital-status category standardized for public dissemination.	Categorical/Text	0	6
ETHNICITY	Student ethnicity represented through standardized public categories.	Categorical/Text	0	6
DISABILITY	Binary disability-status indicator. Detailed disability type and percentage are not included in the public release.	Categorical/Text	0	2

Variable Dictionary - Part II

Variable	Description	Type	Missing	Unique
INSTITUTION_TYPE	Generic institution-type category. Institution-specific names are not included in the public release.	Categorical/Text	0	5
NATIONALITY_COUNTRY	Generalized nationality category. Country-specific nationality values were aggregated into Ecuadorian and International.	Categorical/Text	0	2
TRANSFER_STUDENT_FLAG	Binary indicator identifying verified transfer-student status.	Categorical/Text	0	2
PAYMENT_PLAN	Non-monetary institutional payment-plan category. Complete released categories and frequencies are listed in the financial-variable category/value tables.	Categorical/Text	0	2
ONLINE_ADMISSION_TYPE	Standardized category describing the online admission pathway.	Categorical/Text	0	15
PAYMENT_DEFERRAL_TYPE	Numeric duration of the institutionally recorded payment deferral, expressed in months. The public variable corresponds to the institutional source field TIPO_DIFIRIMIENTO. A value of 0 indicates no payment deferral; positive values indicate the corresponding number of months. Values are quantitative durations rather than nominal category codes or monetary amounts. Complete observed values and frequencies are listed in the financial-variable category/value tables.	Numeric	0	25
SCHOLARSHIP_NAME	Standardized scholarship or discount category. Monetary scholarship amounts are not included. Complete released categories and frequencies are listed in the financial-variable category/value tables.	Categorical/Text	0	9
TARGET_RISK	Binary contemporaneous administrative target derived during study-level data curation from the restricted source variables STUDENT_TYPE and GRADUATION_DATE. TARGET_RISK was not a native field in the restricted institutional research extract. Class 0 indicates alumni status or a valid graduation date; class 1 indicates that the student was not recorded as alumni and no valid graduation date was available. STUDENT_TYPE and GRADUATION_DATE are excluded from the public predictor matrix.	Numeric	0	2
REGION	Generalized geographic region. Public categories are Coastal Region, Highlands Region, Amazon Region, and Insular Region. Province-, canton-, and sector-level information is not included in the public release.	Categorical/Text	0	4

TARGET_RISK Operational Definition

TARGET_RISK = 0: STUDENT_TYPE was Alumni or a valid GRADUATION_DATE was recorded.

TARGET_RISK = 1: the student was not recorded as Alumni and no valid GRADUATION_DATE was recorded.

No grade threshold, academic-performance cutoff, statistical model, weighted risk score, or institutional risk index was used to define TARGET_RISK. STUDENT_TYPE and GRADUATION_DATE were excluded from the public dataset and from the benchmark predictor matrix after target construction.

Documented Public-Release Treatments and Generalization

Variable / field	Public treatment	Rationale
Institutional student-record code	Replaced one-to-one with release-specific PUBLIC_RECORD_ID during public-release preparation. No public crosswalk to the institutional identifier is released.	Preserves within-release record reference while removing the institutional record code; PUBLIC_RECORD_ID is not intended as a longitudinal linkage key.
AGE	Retained as integer age; not converted to age bands.	Preserves analytical utility while direct personal identifiers are absent from the public dataset.
DISABILITY	Only binary Yes/No status retained; detailed disability type and percentage excluded.	Reduces sensitive detail while retaining accessibility-related analytical utility.
NATIONALITY_COUNTRY	Country-specific values generalized to Ecuadorian and International.	Reduces disclosure risk from rare country-specific categories while preserving domestic/international analysis.
REGION	Public representation generalized to Coastal Region, Highlands Region, Amazon Region, and Insular Region. Province-, canton-, and sector-level information is not included in the public release.	Avoids detailed geographic granularity while retaining broad geographic utility.
Financial and payment-related information	The public dataset includes PAYMENT_PLAN, PAYMENT_DEFERRAL_TYPE, and SCHOLARSHIP_NAME. PAYMENT_PLAN and SCHOLARSHIP_NAME are non-monetary categorical descriptors; PAYMENT_DEFERRAL_TYPE records payment-deferral duration in months (0 = no payment deferral). No financial amounts or detailed personal financial records are released.	Documents institutionally recorded payment context without publishing monetary, banking, debt, or income information.
Other institutional / administrative detail	Higher-granularity administrative tracking variables and fields not included in the final public schema are not released.	Data minimization while retaining analytically useful public variables.

Universidad ECOTEC performed institutional preparation and anonymization before delivering the research dataset to the authors. The restricted research extract available to the study team did not include names, national identification numbers, email addresses, or telephone numbers. The public analytical workflows do not perform or reconstruct the upstream institutional anonymization process.

Final Generalized Public Categories

Variable	Category	n
NATIONALITY_COUNTRY	Ecuadorian	12,542
NATIONALITY_COUNTRY	International	90
REGION	Coastal Region	11,063
REGION	Highlands Region	1,338
REGION	Amazon Region	190
REGION	Insular Region	41

Released Financial-Variable Categories and Values - Part I

The tables below report the complete public category/value sets and observed frequencies for the three released financial and payment-related variables. These variables contain no missing values in the finalized public dataset (N = 12,632) and should be interpreted as institutional administrative descriptors rather than as direct measures of individual socioeconomic status.

PAYMENT_PLAN

Public category	n	%
No	6,977	55.23
Yes	5,655	44.77

PAYMENT_DEFERRAL_TYPE

Deferral duration (months)	n	%
0	7,501	59.38
12	12	0.09
18	1	0.01
24	1,137	9.00
25	13	0.10
26	84	0.66
28	2	0.02
30	11	0.09
31	2	0.02
32	1	0.01
34	3	0.02
36	6	0.05
37	6	0.05
38	1	0.01
39	1	0.01
40	6	0.05
41	5	0.04
42	5	0.04
43	8	0.06
44	6	0.05
46	9	0.07
47	9	0.07
48	3,392	26.85
51	1	0.01
60	410	3.25

Interpretation note. PAYMENT_DEFERRAL_TYPE corresponds to the institutional source field TIPO_DIFIRIMIENTO. Released values represent the institutionally recorded payment-deferral duration in months. A value of 0 indicates no payment deferral; positive values indicate the corresponding number of months. These values are quantitative durations, not nominal category codes or monetary amounts. Twenty-five distinct month values are observed in the public release.

Released Financial-Variable Categories and Values - Part II

SCHOLARSHIP_NAME

Public category	n	%
No Scholarship	9,206	72.88
DASE Academic Excellence Scholarship	1,174	9.29
Scholarship/Discount - CRM Migration	877	6.94
Agreement Scholarship	843	6.67
Institutional Scholarship	480	3.80
Institutional Document	22	0.17
Scholarships and Discounts	19	0.15
Support Scholarship	10	0.08
Top High School Graduate Scholarship	1	0.01

Financial-variable interpretation and privacy notes

- PAYMENT_PLAN and SCHOLARSHIP_NAME are non-monetary categorical administrative descriptors; PAYMENT_DEFERRAL_TYPE is a numeric administrative duration expressed in months.
- The public release contains no tuition amounts, transaction histories, banking information, debt balances, personal income information, or monetary scholarship amounts.
- Sparse categories and low-frequency month values should be considered before subgroup or fairness analyses. Scholarship and payment-plan categories, as well as payment-deferral duration, may be associated with socioeconomic or institutional circumstances and may therefore function as proxy variables in some analyses.
- Category/value frequencies are descriptive of this institutional cross-sectional release and should not be interpreted as population-level prevalence outside the dataset.
- The documented public-release treatments are intended to reduce disclosure risk while preserving analytical utility; the release does not claim zero re-identification risk or a formal k-anonymity, l-diversity, or differential-privacy guarantee.